

Actionability Affordance Factorization

Cross-variant follow-up on `actionability_algebra_v3` and `v4`: the fused 4-way permission label is weaker than the primitive affordance bits.

DATE	PHASES	PROMPTS	TEST GROUPS
21 March 2026	v3 and v4	96 per phase	6 held-out groups

MAIN READ

`permission_mode` is probably the wrong probe target. The model appears to represent primitive affordance bits more cleanly than the fused 4-way mode.

V3 CAN BUY	V3 OBSERVE VS ACT	V4 CAN SELL	TOP-SYMBOL INVARIANCE
0.875	0.833	0.750	1.000
<code>constraints_eos L40</code>	<code>last_token L40</code>	<code>constraints_eos L41</code>	unchanged

Why This Matters

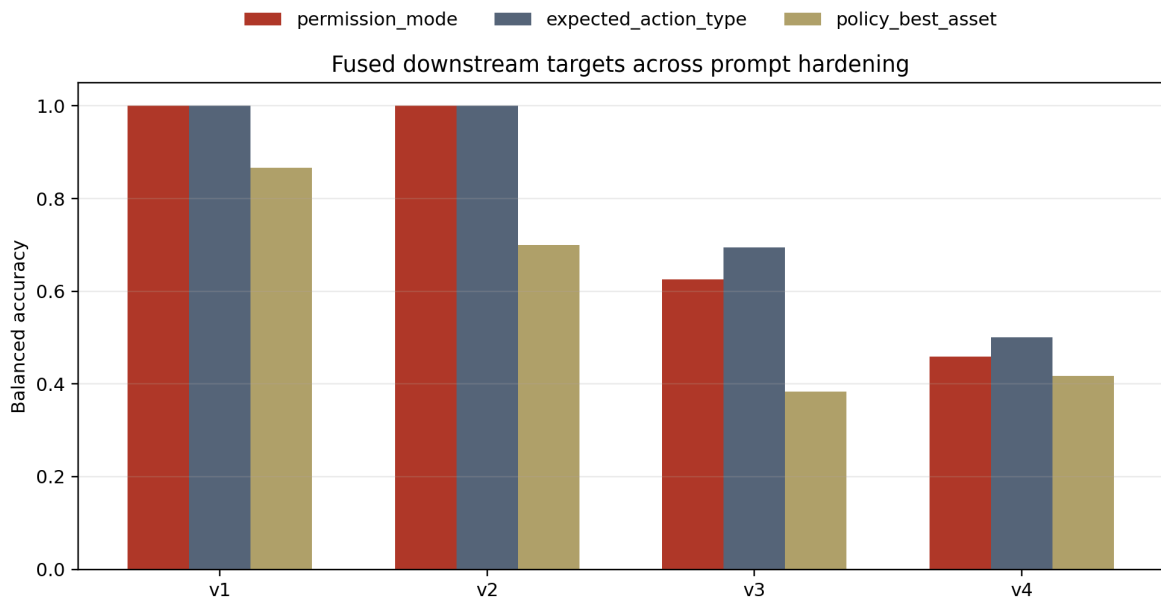
The counting-manifolds paper explicitly warns that early probe attempts can fail when the target is a fused variable rather than the quantity the model actually represents cleanly.

That seems to be happening here.

- The fused 4-way `permission_mode` label is weaker under hardened prompt wording.
- But the primitive bits `can_buy`, `can_sell`, and `observe_vs_act` survive materially better on the same pooled activations in both `v3` and `v4`.

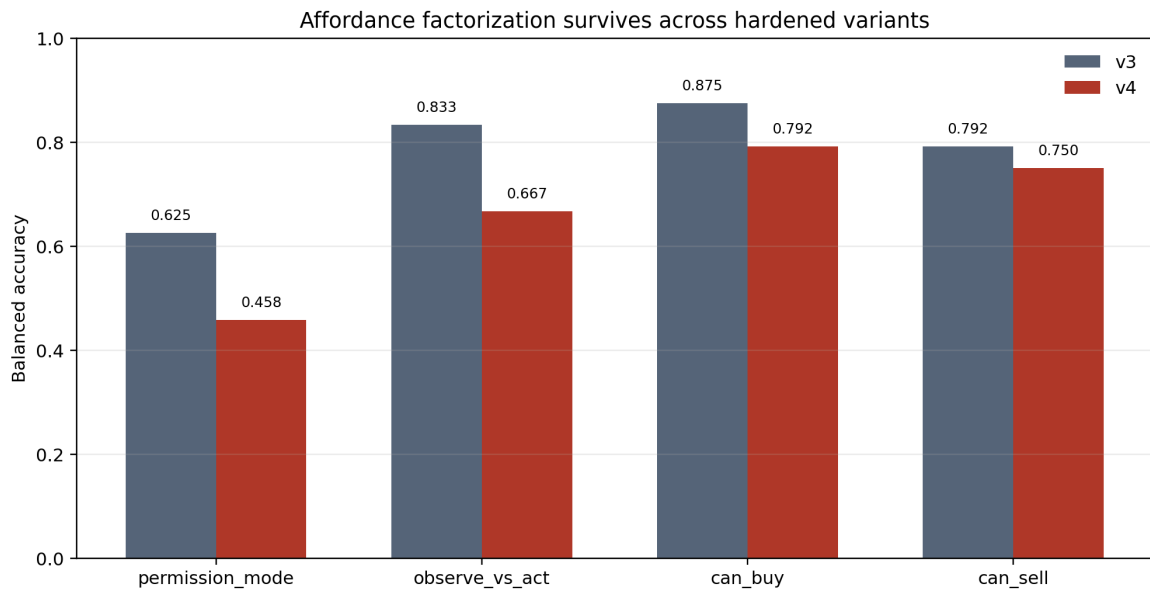
This is the first result in the actionability line that suggests a more precise mechanistic target rather than just a weaker or stronger version of the same story.

Phase Progression



Fused downstream targets across v1 through v4. v2 looked strong, but prompt hardening in v3 and v4 degraded the fused permission-side reads sharply.

v4 Changes The Story

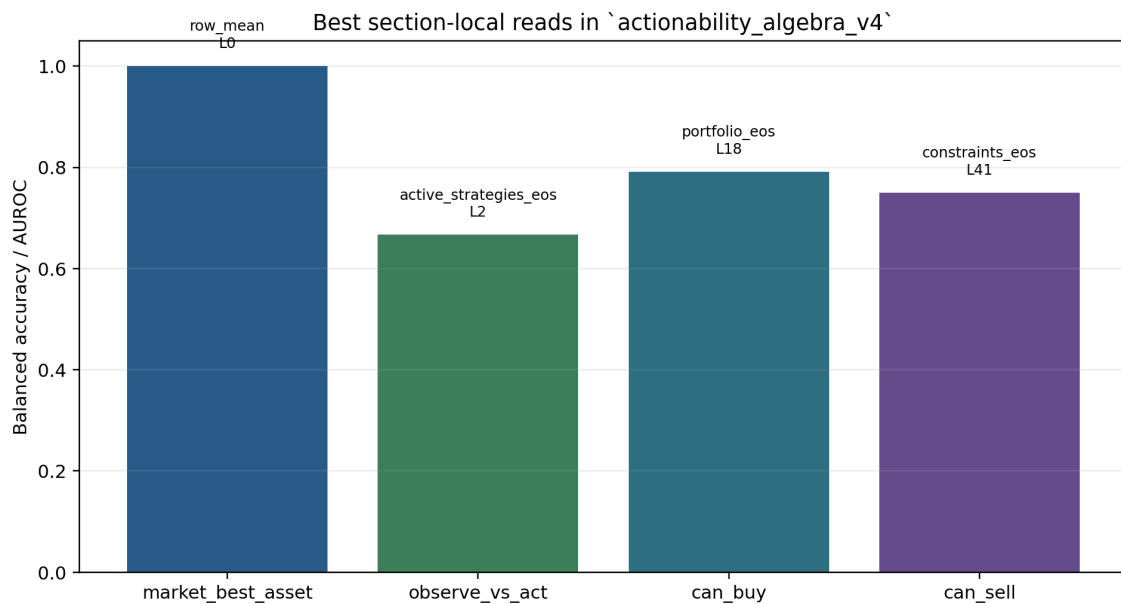


Across both hardened variants, decomposed affordance bits outperform the fused 4-way mode by a large margin.

Target	Best State	Balanced Accuracy
v3 permission_mode	active_strategies_eos L2	0.625
v3 observe_vs_act	last_token L40	0.833

Target	Best State	Balanced Accuracy
v3 can_buy	constraints_eos L40	0.875
v3 can_sell	active_strategies_eos L3	0.792
v4 permission_mode	portfolio_eos L13	0.458
v4 observe_vs_act	active_strategies_eos L2	0.667
v4 can_buy	portfolio_eos L18	0.792
v4 can_sell	constraints_eos L41	0.750

Section Read



v4 still shows the primitive affordance bits landing in plausible different downstream sections. The exact best section moves between v3 and v4, but the factorized affordance story itself remains.

Interpretation

SUPPORTED NOW

The actionability computation looks more factorized than the earlier fused-label framing suggested.

NOT SUPPORTED YET

A clean monolithic permission_mode representation is still not established.

Best current interpretation:

- early market preference is still robust and easy
- downstream permission is not one crisp fused state
- instead, the model appears to carry primitive affordances separately
- that means the next probe family should target `can_buy`, `can_sell`, and `observe_vs_act`, not just `permission_mode`

Next Step

The best next experiment is no longer another fused permission rerun. It is:

- carry the factorized labels into the next synthetic prompt-hardening pass
- test whether `can_buy` and `can_sell` survive additional paraphrase and row-format variation
- then bridge those binary affordance probes into real DX paired reruns

Actionability Affordance Factorization — 21 March 2026. Synthetic v3 and v4 plus decomposed affordance probes on the same pooled residuals.